Bone Scintigraphy
Lec "2"

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Learning Objectives

By the end of this lecture, the technologist should be able to:

- ❑ Explain the principle of bone scintigraphy.
- ❑ Describe radiopharmaceuticals used for bone imaging.
- ❑ Prepare and administer the radiopharmaceutical safely.
- ❑ Perform the three phases of bone scintigraphy.
- ❑ Understand patient preparation and imaging protocols.
- ❑ Recognize normal skeletal distribution.
- ❑ Identify common abnormal patterns.
- ❑ Understand SPECT/SPECT-CT applications.





Patient preparation before delayed imaging



During the uptake period:

- ❑ Encourage hydration
- ❑ The patient should drink fluids unless medically contraindicated.
- ❑ Encourage frequent voiding
- ❑ The patient should void immediately before imaging.

This:

- ❑ Reduces urinary bladder radiation dose
- ❑ Reduces background activity
- ❑ Improves pelvic image quality

Delayed Whole-Body Acquisition

For many indications, delayed imaging includes anterior and posterior whole-body images.



Spot Views

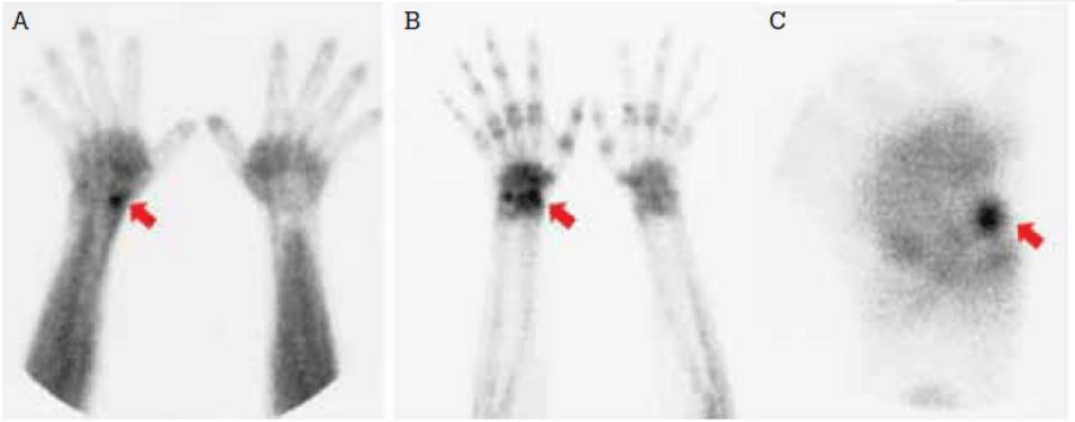
Additional static images should be obtained when an abnormality is:

- ❑ Poorly characterized on whole-body imaging
- ❑ Very small
- ❑ Located in a complex anatomical region suspected to represent a fracture or infection

Possible views include:

Anterior Posterior
Lateral Oblique

For example, a suspicious rib lesion on whole-body imaging may require oblique views.



Scaphoid fracture with corresponding intense localised blood pool (A) and delayed phase activity (B). Magnified (pinhole) delayed image (C)

SPECT / SPECT-CT

SPECT or SPECT/CT can be performed during the delayed phase.

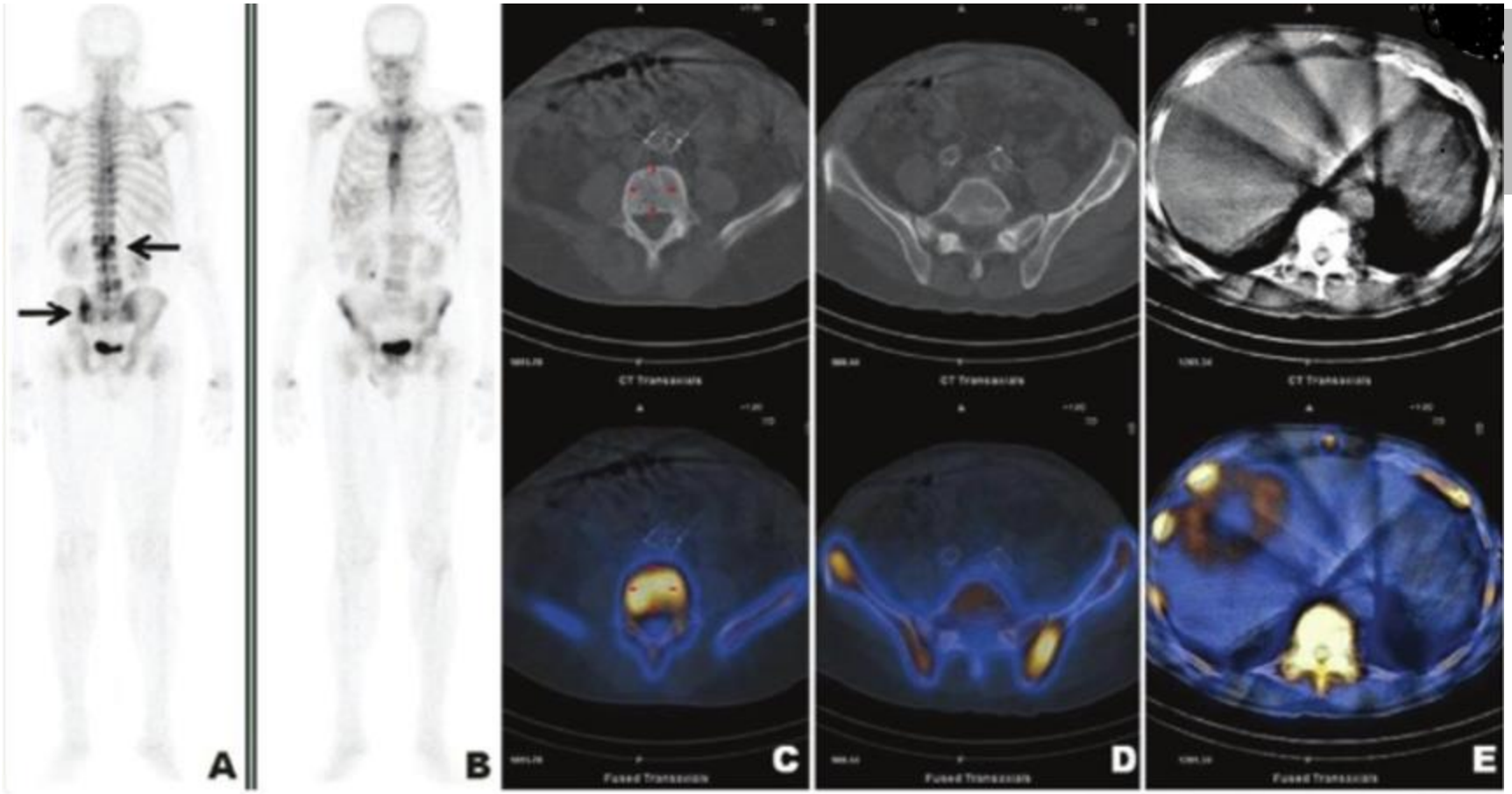
This is particularly valuable when planar imaging demonstrates an equivocal lesion.

SPECT advantages

Provides:

- ❑ Tomographic localization
- ❑ Improved contrast
- ❑ Better lesion conspicuity
- ❑ Separation of overlapping structures





Planar whole body bone scan images (A & B) show bone metastasis in the sternum, L2, L3, left ilium. SPECT/CT localizes bone metastasis (C & D) and finds unsuspected liver metastasis (E) from a previously unknown synchronous lung cancer.



SPECT/CT advantages

Combines:

Functional information from SPECT + anatomical information from CT. This is especially useful for:

- ❑ Spine
- ❑ Pelvis
- ❑ Sternum
- ❑ Ribs
- ❑ Sacroiliac joints
- ❑ Complex joints
- ❑ Postoperative anatomy
- ❑ Prosthetic joints



A “Three-phase“ examination will provide valuable additional information. This involves a dynamic flow study of the area of interest with rapid sequential images taken every 2-3 s for 30 s.

This is followed by a blood pool image at 1 min, when the radiopharmaceutical is still predominantly within the vascular compartment.

Delayed static images are then obtained 2-4 h and, rarely, 24 h later as dictated by the clinical situation and by the workload of the department.



Normal Appearances of bone scintigraphy

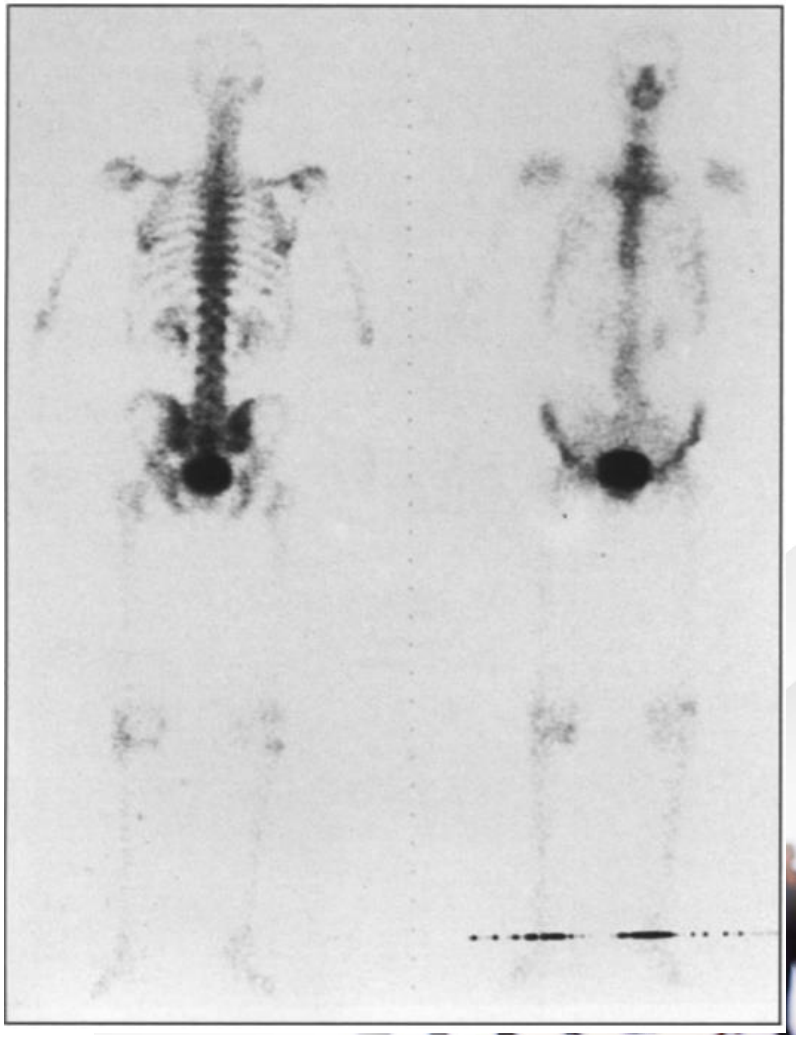




Normal Appearances

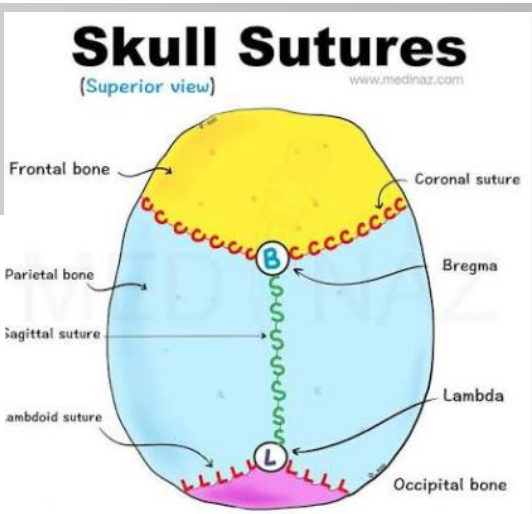
The most important single feature in a normal skeletal scintigram is symmetry about the midline in the sagittal plane.
Thus the left and right halves of the body are virtually mirror images of each other.

The count rate is greatest in those parts of the skeleton subjected to the greatest stress, and therefore undergoing the fastest turnover (e.g. the sacroiliac region, where the weight of the body is transferrred from the trunk to the pelvis. or the inferior angles of the scapulae. where the balance of muscular forces can create considerable tension).
There is also relatively high uptake along the spine and around the hips and other joints.

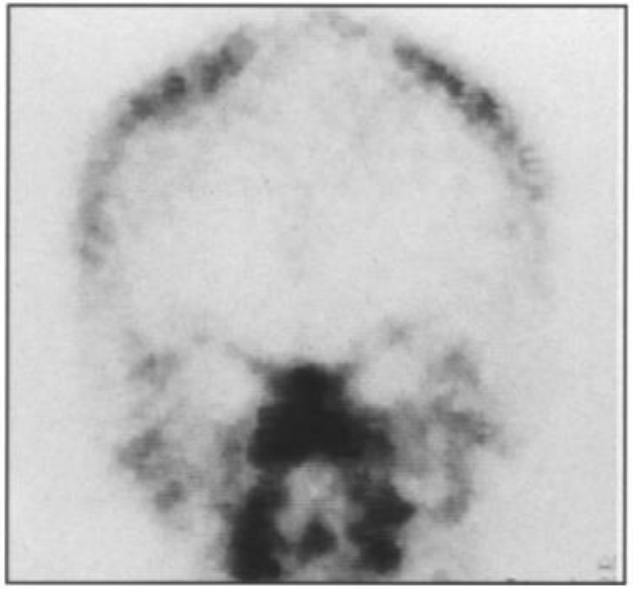


Head

The count rate detected from the skull is comparatively low. Variations in uptake within the skull may be seen. especially in relation to the suture lines.



In the anterior projection the orbits and facial bones are clearly visualised. A higher count rate is observed from the facial bones. especially the maxilla and ethmoids .where the stresses resulting from mastication are transmitted to the facial skeleton.



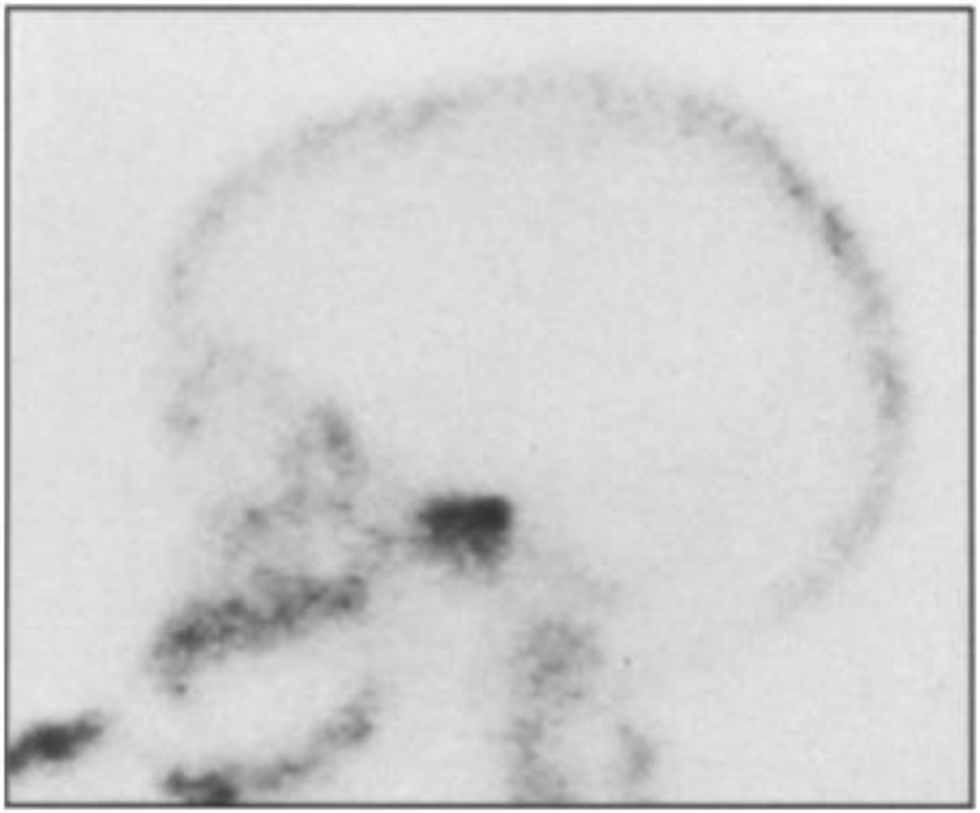
Anterior skull projection. The uptake in the ethmoids and maxilla is a normal feature.

Lateral view of adult skull showing uptake in suture lines.

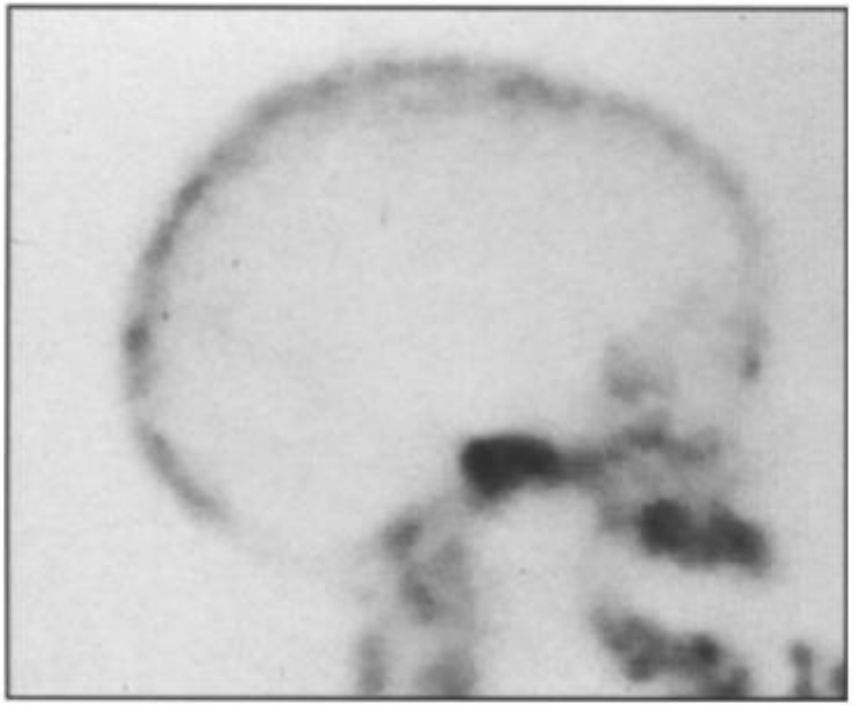


In older children and adolescents difficulty may arise if it is not appreciated that there is an important suture. which can be very active. between the sphenoid and the basiocciput . This is a normal feature and should not be confused with sphenoidal sinusitis or pathology in the region of the clivus.

Lateral projection of the skull of a 7-year-old child. The high uptake in the base is due to the suture between the sphenoid and the basiocciput. which was in process of fusion.
The relatively high uptake in the maxilla compared with the mandible. This reflects the state of dentition.



During the eruption of dentition uptake in the maxilla and mandible may be uneven.



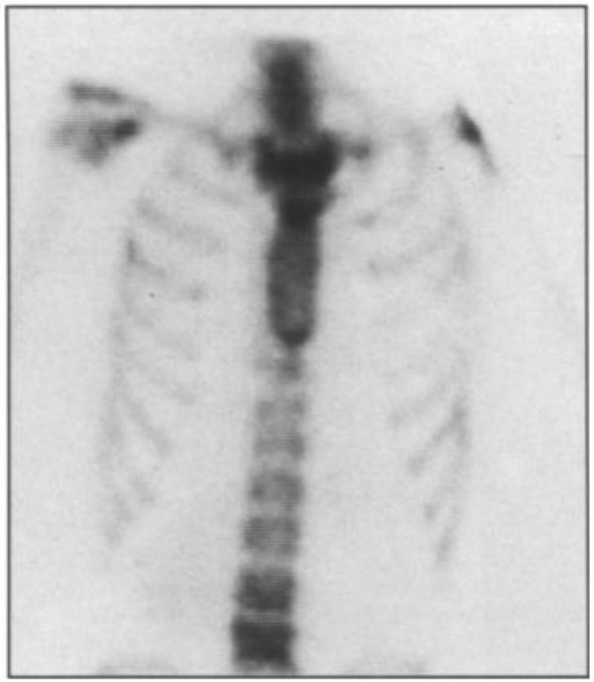
Lateral skull of a l0-year-old child. The suture between the sphenoid and the basiocciput is much more active and there is much more asymmetry in the maxilla because of the eruption of various teeth.

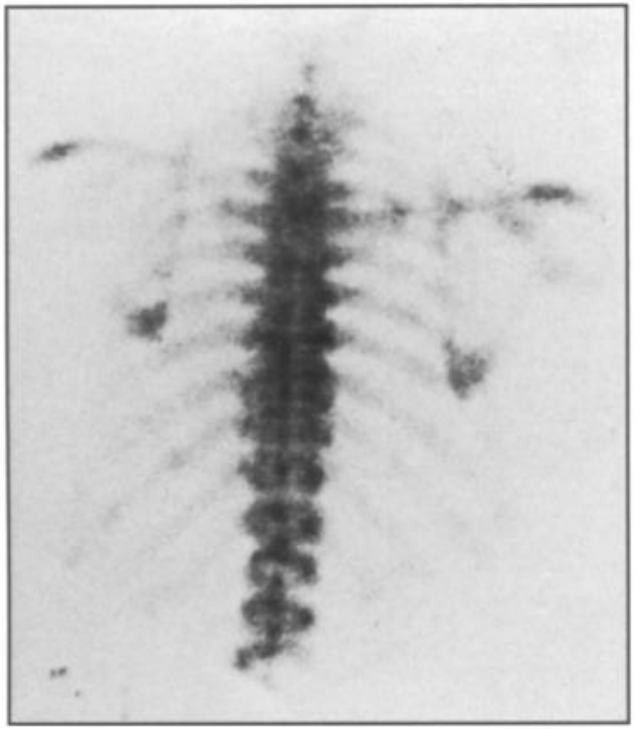


Neck

The spinous process of the vertebra prominens (C-7) is usually visible, distinct and may have a higher count rate than adjacent bones.

Uptake in the neck in the absence of free pertechnetate is usually in the thyroid cartilage (which is not necessarily calcified radiologically) or hyoid rather than thyroid gland.

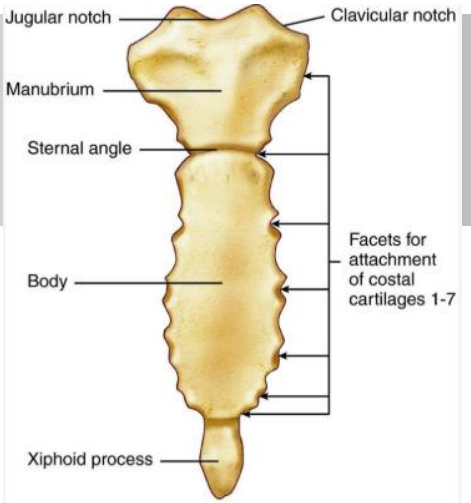


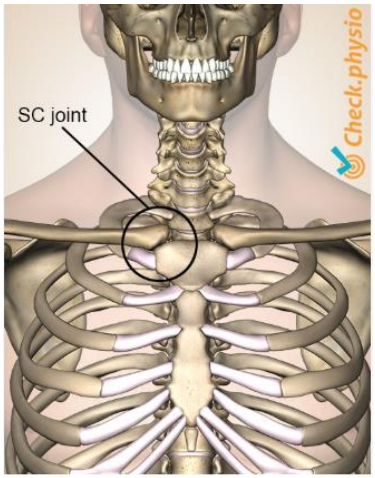


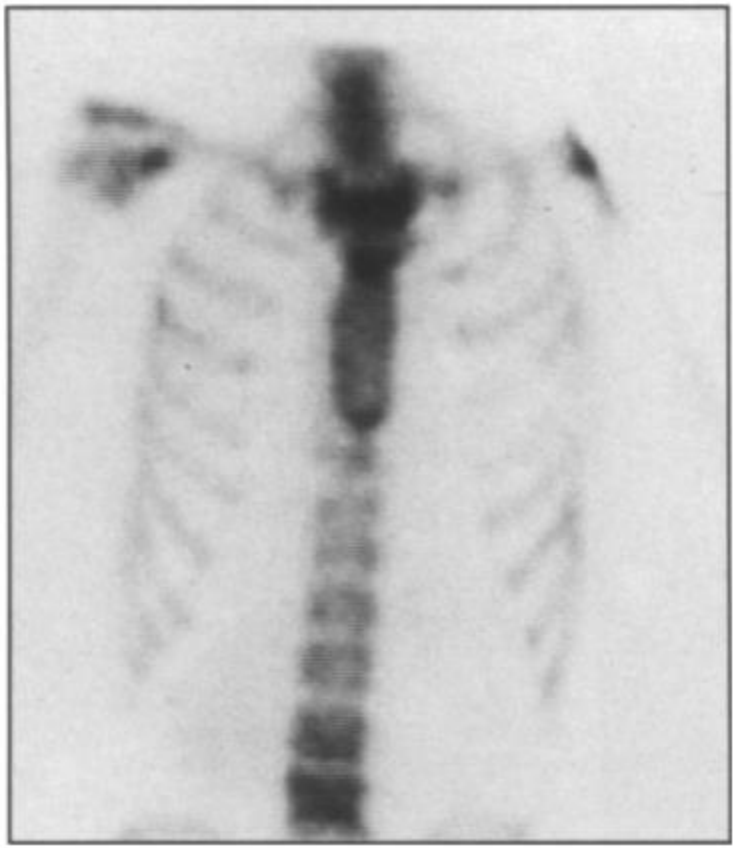
Thorax

In the anterior projection of the thorax the sternum is clearly seen and particularly in older subjects. The thyroid cartilage may be identified above it.

The sternomanubrial joint is often relatively active. particularly in older subjects and in occasional individuals there is high tracer uptake at the sternoclavicular joints.







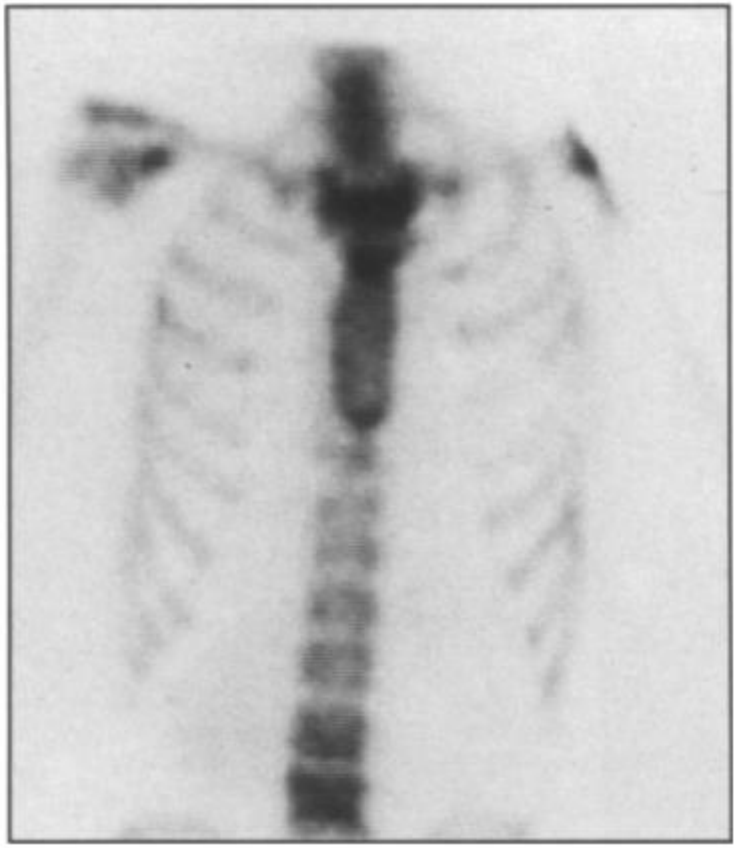
Thorax

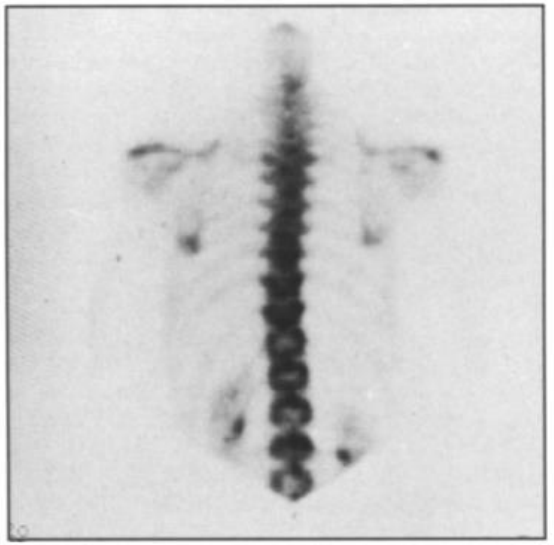
The clavicles are not well seen. but the acromioclavicular joints often exhibit relatively high but usually symmetrical uptake.

Calcified costal cartilages are often visualised in older subjects.

All of the ribs should be visible in the posterior projection of the thorax.

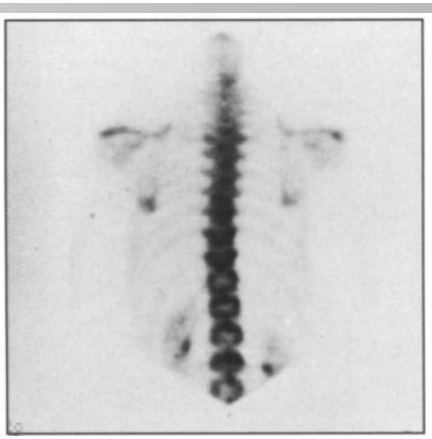
The scapulae are well seen. uptake being mainly concentrated at the angles. where muscle stresses are greatest. Where they overlie ribs.





Spine

The upper thoracic vertebrae cannot be individually identified, but there should be no difficulty in visualising the disc spaces between the lower ones.



In the normal subject lower thoracic and lumbar pedicles and spinous processes are readily identified

The normal lumbar lordosis often makes the lower lumbar vertebrae appear prominent on anterior views of the Spine.

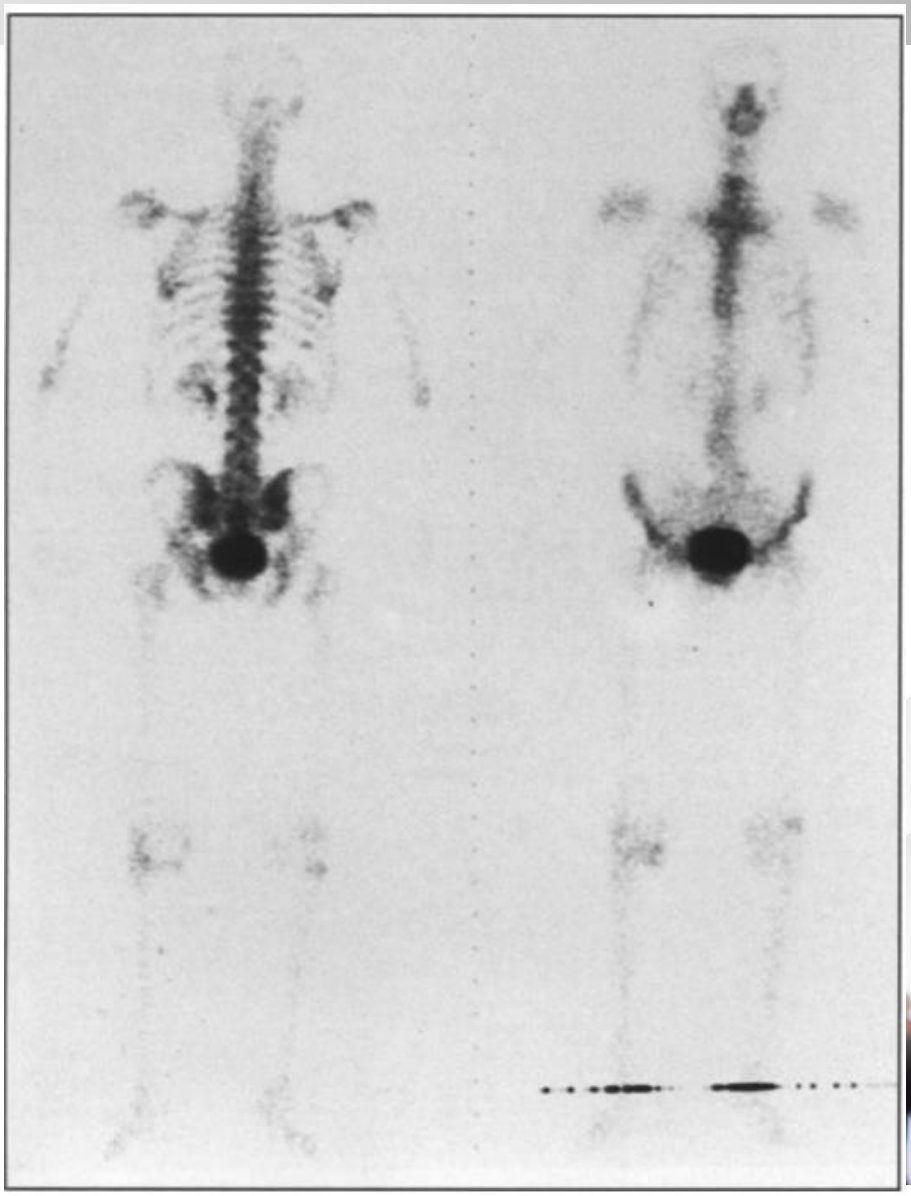


The kidneys are seen in views of the lumbar spine, but the collecting system and ureters are not visualised at all unless there is urinary tract outflow obstruction.

Pelvis

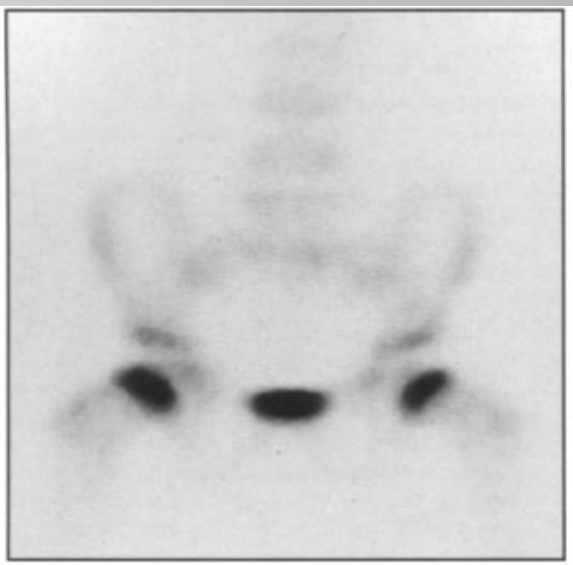
There is often some residual activity in the bladder, the apparent position and shape of which differs in the anterior and posterior projections and may also change between supine, prone and erect.

The symphysis and adjacent parts of the pubis and ischium are consequently often obscured.

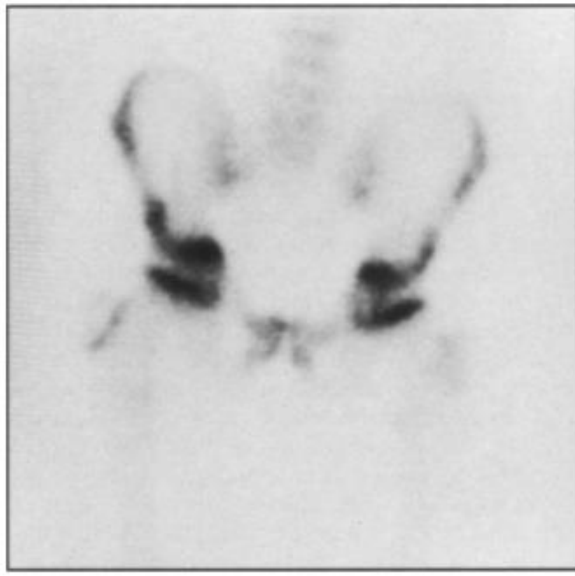


Limbs

The normal femoral heads are seen only faintly in the adult, although the region of the greater trochanter and the intertrochanteric ridge can usually be identified , In children the epiphysis for the greater trochanter as well as that for the head of the femur are readily visible.

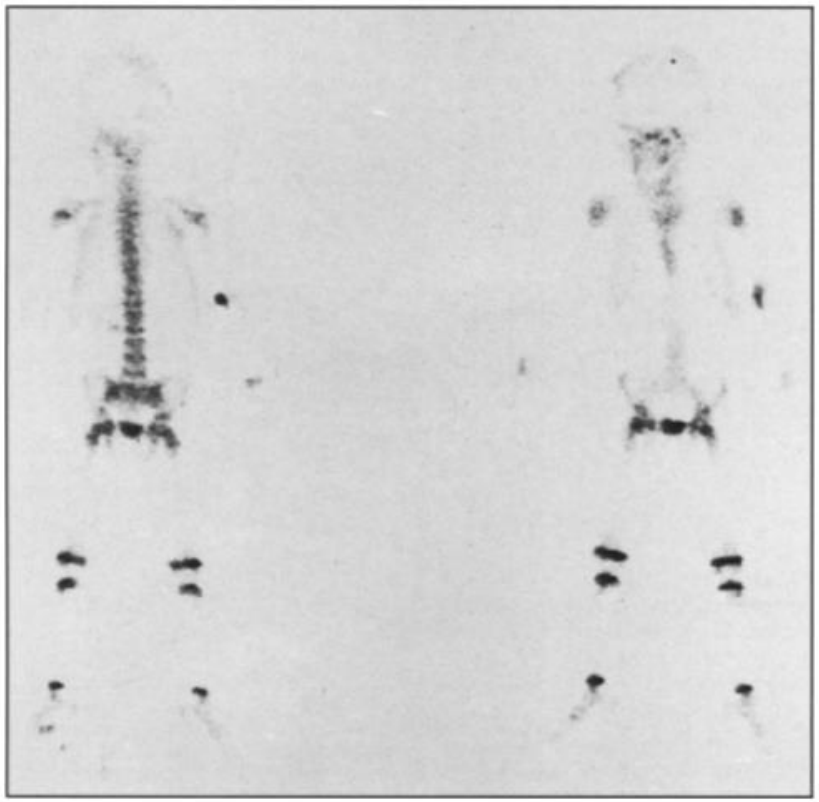


The shafts of the long bones are virtually invisible at intensity settings suitable for displaying the rest of the skeleton, but uptake can just be seen around the shoulders. elbows, knees, ankles and feet are sometimes visible and are commonly asymmetrical as turnover is related to local stress.



In childhood, growth occurs principally in the epiphyses adjacent to the knee and away from the elbows.
The activity of the epiphyses varies considerably with age.
The age at which epiphyses fuse radiographically is well documented. Most epiphyses appear to become more active shortly before fusion.





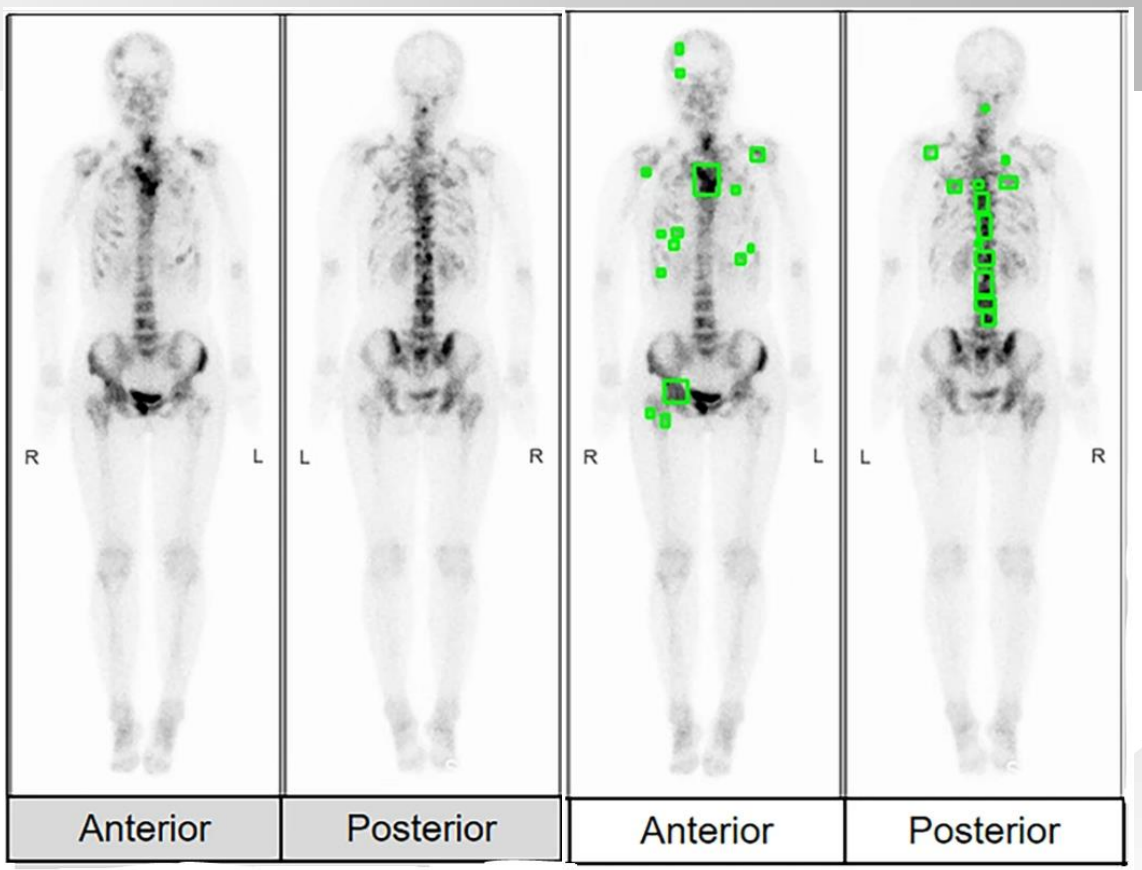
Indications

Oncology

1. Detection of skeletal metastases
2. Assessment of suspected bone involvement
3. Evaluation of treatment response in selected circumstances

Common primary tumors associated with bone metastases *include:*

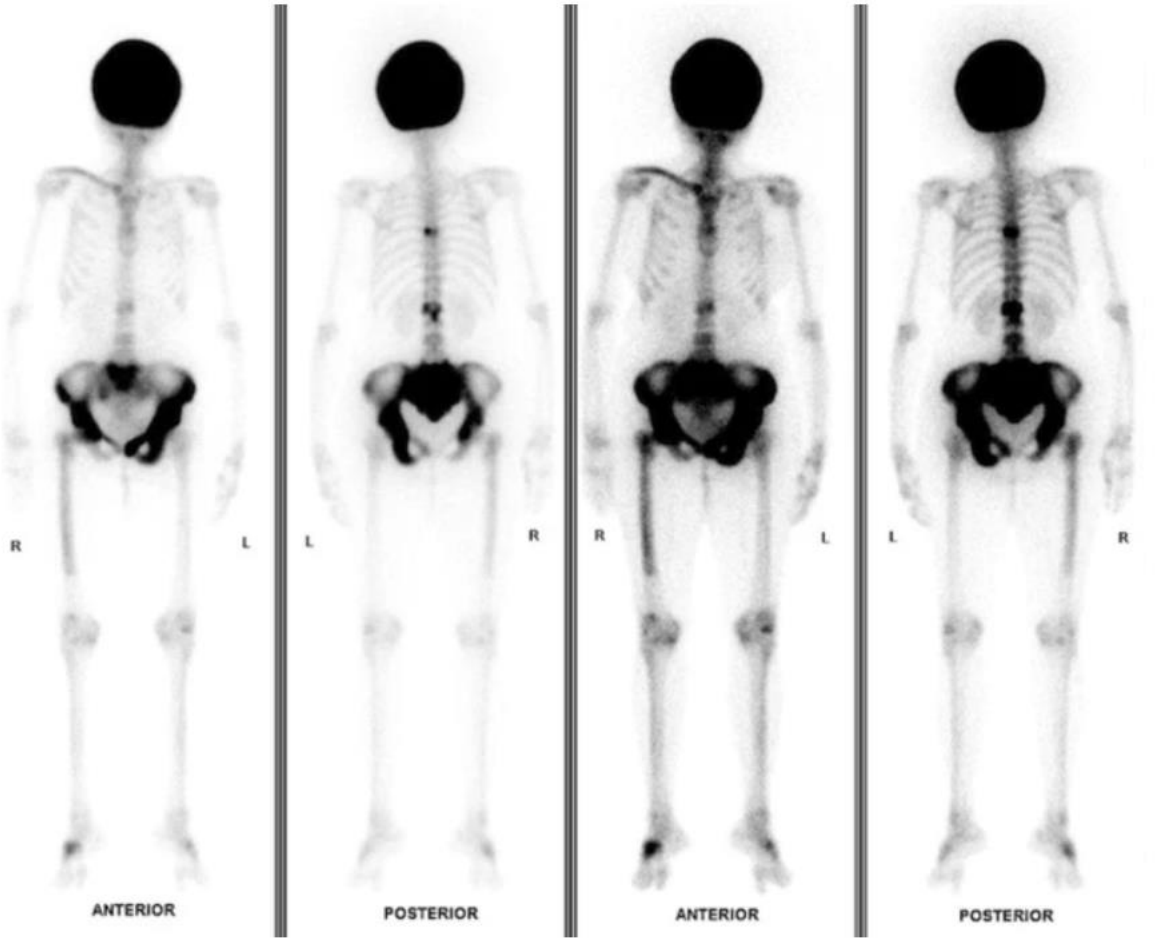
- ❑ Breast cancer
- ❑ Prostate cancer
- ❑ Lung cancer
- ❑ Renal cancer
- ❑ Thyroid cancer





Non-oncologic indications

Occult fractures
Stress fractures
Osteomyelitis
Prosthetic joint complications
Avascular necrosis
Paget disease
Arthropathies
Sports injuries
Unexplained bone pain



Bone scan at baseline showing uptake in skull, vertebrae and pelvis suggestive of Paget's disease of bone
'Mickey Mouse' sign of Paget's

Osteoblastic metastases

Usually produce:

Increased tracer uptake → "hot lesion"

Classically associated with:

Prostate carcinoma

Some breast cancers

Neuroendocrine tumors

Osteolytic lesions

May show:

Normal uptake

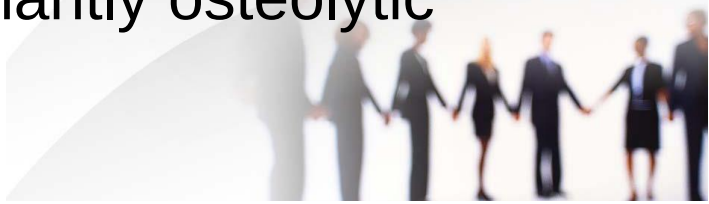
Reduced uptake

Mild uptake

Increased uptake if there is a reactive osteoblastic component

Therefore:

A negative bone scan does not exclude all bone metastases. This is particularly important for predominantly osteolytic tumors.



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Thank You



